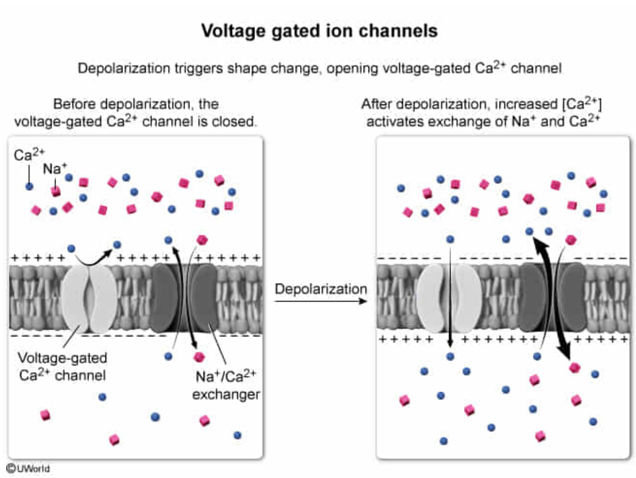
After cardiac cells depolarize under physiologically normal conditions, extracellular Na^+ enters the cells via a specific plasma membrane protein. The Na^+ transport provides the energy needed to pump intracellular Ca^{2+} to the extracellular space. This Na^+/Ca^{2+} exchange depends on a preceding increase in intracellular [Ca^{2+}] in response to the cell's decreased electric potential. This preceding increase in Ca^{2+} is most likely evidence of:

- A. direct voltage-dependent activation of the exchanger. [4%]
- B. entry of Ca^{2+} ions into the intracellular fluid through a voltage-dependent membrane channel. [56%]
- C. activation of a ligand-gated Ca^{2+} channel. [10%]
- D. transport of Ca^{2+} ions against their concentration gradient through a voltage-gated membrane channel. [27%]

Correct
56% answered correctly

Explanation:



Solutes can cross cellular membranes by various means, depending on the characteristics (eg, size, charge, polarity) of the solute(s) being transported and the environments (eg, charge, concentration of various solutes) on each side of the membrane. Because they are charged, ions are hydrophilic and require a membrane protein to cross the hydrophobic interior of the cell membrane.

The question describes the exchange of two ions (Na^+ and Ca^{2+}) across the plasma membrane of cardiac cells by a membrane protein (ie, the protein is an antiporter). This Na^+/Ca^{2+} **exchange** occurs **after depolarization**, which causes an **increase** in the **intracellular Ca^{2+}** concentration. Because depolarization is a reduction in membrane potential (ie, voltage), this information **suggests** that **depolarization promotes entry** of Ca^{2+} ions **through a voltage-gated** membrane **channel** (ie, a channel that **opens in response to a change in membrane potential**).

(Choice A) The dependence of the Na^{+}/Ca^{2+} exchange on a prior increase in intracellular Ca^{2+} suggests that the two processes are separate, and that the voltage change initially causes net Ca^{2+} *entry* into the cell rather than net Ca^{2+} exit out of the cell.

(Choice C) A ligand-gated Ca^{2+} channel would require binding of another molecule to operate, and no such molecule is mentioned. The question states that Ca^{2+} entry occurs in response to depolarization, suggesting a voltage-gated channel, not a ligand-gated channel.

(Choice D) The question indicates that Ca^{2+} transport from intracellular to extracellular space requires energy input; therefore, Na^{+}/Ca^{2+} exchange involves transport *against* the calcium gradient. However, the voltage-gated channel facilitates Ca^{2+} *entry* into the cell, which must occur *down* the gradient.

Educational objective:
Voltage-gated ion channels transport ions in response to changes in membrane potential.

Time spent:0 sec
Subject:
Biochemistry

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System:
Amino Acids and Proteins